ONE-PAGE BRIEF

Post-quantum cryptography review: rs-soroban-env @ 0c99950e9612

Client: Stellar Development Foundation — public repository (passive review) · Source: github.com/stellar/rs-soroban-env, commit `0c99950e9612` · Reviewed: 2026-11-06 16:05 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY

Level 2 of 4

post-quantum in use outside the critical paths

FINDINGS

0 Critical **0** High **10** Medium **6** Low **2** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Extend post-quantum protection to the critical roles that stay classical, starting with what can be recorded today (wrapped keys, channel key exchange).

WHAT MATTERS MOST

Medium	QSI-2026-19557	**Quantum-vulnerable Ed25519: transaction and message signing (2 file(s))** preliminary What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium	QSI-2026-19561	**Quantum-vulnerable ECDSA: transaction and message signing (2 file(s))** preliminary What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium	QSI-2026-19558	**Quantum-vulnerable Ed25519: role not determined (9 file(s))** preliminary What to do: Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form...
Medium	QSI-2026-19559	**Quantum-vulnerable BLS12-381: role not determined (5 file(s))** preliminary What to do: Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an abstraction layer for...

12 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium transaction and message signing	ECDSA, Ed25519	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum...
Medium authentication and identity (certificates, tokens, device attestation)	Ed25519	Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them
Medium role not determined	Ed25519, ECDSA, BLS12-381, BN254 (alt_bn128)	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

KNOWN VULNERABLE COMPONENTS

Advisories for the versions in use: 9; unlikely to apply to this build: 2 (a Cargo feature that is off, affected functions not named in the code, or a platform that is not built); to review: 7.

Static review of source code and manifests; files analysed: 256. No requests were sent to the owner's systems.

Full report, content SHA-256: `9b85f3462552f82353205b89d9274711efee1e99748e272a6e7abd94e0172d1e`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.